Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5 HT	(a)			Within the ranges 45 to 90 (63) (1) 1.75 to 2.35 (1.9) (1) -20 to -200 (–101) (1)			3		3	
	(b)			**Indicative content** Similar trends in terms of density and atomic radius. Alkali metals are all solids and the melting point/boiling point decreases as they get bigger. Halogens can be solids (I_2 and At_2), liquids (Br_2), or gases (F_2 and Cl_2) at room temperature and the melting point/boiling point increases as they get bigger. Reactivity increases down the group for the alkali metals but decreases down the group for the halogens. Alkali metals have 1 electron in the outer shell but halogens have 7 electrons in the outer shell **5-6 marks** Comprehensive comparison of similarities and differences in trends. Expect all properties in tables to be included together with comment about trends in reactivity and electronic structure. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** Comparison of similarities and differences in trends. Expect 3 of the properties in tables to be included and maybe comments about trends in reactivity. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar. inaccuracies in spelling, punctuation and grammar.*	3	3				

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Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Limited comparisons made. May include similarities and/or differences. OR Description of trends without comparison of alkali metals and halogens. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** No attempt made or no response worthy of credit.						
	(c)			Outer shell is further away (from nucleus) / radius of atom gets larger (1) and is more difficult to attract another electron (1) so is <u>less</u> reactive (1)	3					
	(d)			Fluorine and sodium			1			
				Question 5 total	**6**	**3**	**4**	**13**	**3**	

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